

Occupational Safety and Health Engineering

MODULE-01

Outline the purpose of occupational Safety and Health Act of India.

- * Establish Mandatory Occupational Safety and Health Standards
 - * Effective Enforcement of Safety Regulations
 - * Uniform Record Keeping and Reporting
 - * Define Rights and Responsibilities
 - * Establishment of Safety Organizations
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Explain the history of Safety development.

1. Early Industrial Period
2. Formation of National Safety Organizations
3. Development During the Industrial Revolution
4. Introduction of Safety Legislation
5. Development of Safety Programs
6. Growth of Modern Safety Management
7. Three E's of Safety

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1802 → Health and Morals of Apprentices Act

1867 → Factory Inspection

1869 → Mine Safety Act & BLS

1877 → Machinery Guarding Law

1892 → First Safety Program (Joliet Plant)

1913 → National Safety Council Formation

3. Outline the circumstances that lead to development of OSHA and state the mission and purpose of OSHA.

Circumstances that Led to the Development of OSHA

1. High Number of Industrial Accidents
2. Lack of Worker Protection
3. Poor Working Conditions
4. Public Demand for Safety
5. . Development of Safety Programs
6. . Need for Uniform Safety Standards

Mission of OSHA- "To ensure, so far as possible, every working man and woman safe and healthful working conditions and to preserve human resources."

Purpose of OSHA:

- * Establish Mandatory Occupational Safety and Health Standards
 - * Effective Enforcement of Safety Regulations
 - * Uniform Record Keeping and Reporting
 - * Define Rights and Responsibilities
 - * Establishment of Safety Organizations
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Discuss in detail the two responsibilities of employer according to OSHA.

.1 General Responsibility (General Duty Clause)

- Identify workplace hazards.
- Eliminate or control unsafe conditions.
- Provide safe tools, equipment, and machinery.
- Ensure proper ventilation, sanitation, and cleanliness.
- Protect workers from physical, chemical, biological, and ergonomic hazards.
- Promote a safety culture throughout the organization.

2. Specific Responsibility (Compliance with OSHA Standards)

- a) Follow OSHA Standards
 - b) Training and Education
 - c) Record Keeping and Reporting
 - d) Personal Protective Equipment (PPE)
 - e) Workplace Inspections
 - f) Emergency Preparedness
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5. *Enumerate the axioms of Industrial safety.*

1. Accidents are Caused
 2. Accidents can be prevented
 3. Unsafe Acts and Unsafe Conditions Cause Accidents
 4. Management is Responsible for Safety
 5. Safety is a Line Function
 6. Training is Essential for Safety
 7. Safety Improves Productivity
 8. Employee Participation is Necessary
 9. Accident Prevention is Economical
 10. Continuous Improvement is Required
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Heinrich's Axioms of Industrial Safety

1. Injuries result from a completed series of factors, one of which is the accident itself.
 2. An accident can occur only as a result of an unsafe act by a person and a physical or mechanical hazard.
 3. Most accidents are the result of unsafe behavior by people.
 4. An unsafe act or unsafe condition does not always immediately result in an accident or injury.
 5. The reasons why people commit unsafe acts serve as helpful guides in selecting corrective actions.
 6. The severity of accidents is largely by chance and accidents are preventable.
 7. The best accident prevention techniques are analogous with best quality and productivity techniques.
 8. Management should assume responsibility for safety, since it is in the best position to get results.
 9. The supervisor is the key person in the prevention of industrial accidents.
 10. In addition to direct costs (compensation, medical costs), there are also hidden and indirect costs of accidents.
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Briefly explain the Dominos theory of accident causation

10 axioms, Five Dominos in Heinrich's Theory: **[FACTORS]**

- *1. Ancestry and Social Environment*
 - *2. Fault of Person*
 - *3. Unsafe Act and/or Unsafe Condition*
 - *4. Accident*
 - *5. Injury*
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Accident : An unplanned and undesired event resulting in injury, damage, or loss.

Hazard : A source or situation with the potential to cause harm.

Risk : The probability that a hazard will cause harm and the severity of its consequences.

1. ***Engineering***: Engineering involves designing and maintaining workplaces, machines, and equipment in a safe manner to eliminate or reduce hazards.

2. ***Education***: Education refers to providing workers and supervisors with safety knowledge and training

3. ***Enforcement***: Enforcement means ensuring that safety rules, regulations, and procedures are strictly followed.

Justification of Accident Investigation

1. ***Identifies Causes of Accidents***

2. ***Prevents Recurrence***

3. ***Improves Workplace Safety***

4. ***Reduces Losses***

5. ***Legal Compliance***

6. ***Improves Employee Morale***

7. ***Provides Valuable Records***

Methods of Accident Investigation

1. *Observation Method*
 2. *Interview Method*
 3. *Document Review Method*
 4. *Physical Evidence Analysis*
 5. *Root Cause Analysis*
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10. Explain the necessity of Occupational Safety and Health at a work place.

1. *Prevention of Accidents and Injuries*
 2. *Protection of Employee Health*
 3. *Improvement of Productivity*
 4. *Reduction of Financial Losses*
 5. *Legal Compliance*
 6. *Improvement of Employee Morale*
 7. *Protection of Equipment and Property*
 8. *Reduction of Absenteeism*
 9. *Development of Safety Culture*
 10. *Preservation of Human Resources*
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Goals of the National Policy

1. *Reduction in Occupational Accidents and Injuries*
 2. *Prevention of Occupational Diseases*
 3. *Improvement of Workplace Environment*
 4. *Protection of Human Resources*
 5. *Promotion of Safety Culture*
 6. *Sustainable Industrial Development*
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Objectives of the National Policy

1. *Create Awareness*
 2. *Develop Safety Management Systems*
 3. *Strengthen Legal Framework*
 4. *Promote Research and Development*
 5. *Training and Education*
 6. *Improve Working Conditions*
 7. *Control Workplace Hazards*
 8. *Encourage Employee Participation*
 9. *Maintain Accident Records and Statistics*
 10. *Protect the Environment*
 11. *Continuous Improvement*
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Role of Government in Right-to-Know Laws

1. Formulation of Laws and Regulations
 2. Enforcement of Regulations
 3. Collection and Maintenance of Information
 4. Protection of Workers' Rights
 5. Public Awareness and Education
 6. Emergency Planning and Response
 7. Monitoring Environmental Safety
 8. Penalties for Non-Compliance
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Advantages of Right-to-Know Laws

1. Increases Employee Awareness
2. Improves Workplace Safety
3. Prevents Occupational Diseases
4. Promotes Emergency Preparedness.
5. Encourages Safe Handling and Storage
6. Enhances Community Protection
7. Improves Environmental Protection

Disadvantages of Right-to-Know Laws

1. Increased Administrative Burden
 2. Higher Compliance Costs
 3. Information Overload
 4. Disclosure of Confidential Information
 5. Time-Consuming Process
 6. Misinterpretation of Information
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MOD 2

OSHA's Ergonomics Guidelines

1. Worksite Analysis
 2. Risk Factors
 - a. Common CTD Risk Factors
 - b. Back Disorder Risk Factors
 3. Hazard Prevention & Control
 - a. Engineering Controls
 - b. Administrative Controls
 4. PPE
 5. Comprehensive Ergonomics Program
 - a. Management commitment.
 - b. Worker involvement.
 - c. Ergonomics team or committee.
 - d. Training and education.
 - e. Job evaluation.
 - f. Hazard prevention and reduction.
 - g. Early detection and treatment of injuries.
 - h. Medical management of cases.
 - i. Reporting system for ergonomic problems.
 - j. Continuous evaluation and improvement
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Work Analysis Programme for Ergonomics

1. Gathering Information from Available Sources
 - a) Records Analysis and Tracking
 - b) Incidence Rate Calculation
 2. Conducting Baseline Screening Surveys
 - a. Common CTD Risk Factors
 - b. Back Disorder Risk Factors
 3. Ergonomic Job Hazard Analysis
 4. Periodic Surveys, Feedback and Follow-Up Studies
 - a) Feedback and Follow-Up
 - b) Trend Analysis
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Application of Ergonomics for Safety and Health

- A. Improving Worker Safety
 - B. Prevention of Musculoskeletal Disorders (MSDs)
 - C. Better Workplace Design
 - D. Reduction of Fatigue and Stress
 - E. Improvement in Productivity
 - F. Protection of Worker Health
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Constituents of Ergonomics

- 1. Physical Ergonomics
 - 2. Cognitive Ergonomics
 - 3. Organizational Ergonomics
 - 4. Visual Ergonomics
 - 5. Workplace Ergonomics
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Steps in Event Tree Analysis

- 1. Identify the Initiating Event
 - 2. Identify Safety Systems
 - 3. Construct the Event Tree
 - 4. Evaluate Possible Outcomes
 - 5. Assess Risk
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Applications of Event Tree Analysis

- Process industries
 - Chemical plants
 - Nuclear power plants
 - Fire safety analysis
 - Emergency planning
 - Risk assessment studies
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Advantages of ETA	Limitations of ETA
1. Identifies possible accident sequences.	1. Can become complex for large systems.
2. Evaluates effectiveness of safety systems.	2. Requires accurate probability data.
3. Helps in emergency planning.	3. Time-consuming for detailed analysis.
4. Supports risk management.	4. May not identify all possible initiating events.
5. Easy to understand and visualize.	

Procedure for Conducting Ergonomic Job Analysis

1. Identify Jobs with Ergonomic Risk Factors
 2. Perform Workstation Analysis
 3. Evaluate Tools and Equipment
 4. Assess Manual Material Handling and Lifting Hazards
 5. Use Videotape Analysis Method
 6. Verify Risk Factors
 7. Recommend and Implement Control Measures
 8. Conduct Follow-Up Evaluation
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Visual Ergonomics

1. Eye-to-Screen Distance
2. Vertical Location
3. Monitor Tilt
4. Lighting
5. Screen Colours

Benefits of Visual Ergonomics

- Reduces eye fatigue and strain.
 - Improves visibility.
 - Enhances work accuracy.
 - Increases productivity.
 - Prevents headaches and visual discomfort.
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Hazard Prevention and Control Methods

1. Eliminate the Hazard (Engineering Controls)
 2. Improve Work Policies and Procedures (Administrative Controls)
 3. Provide Personal Protective Equipment (PPE)
 4. Establish a Comprehensive Ergonomics Program
 5. Hazard Analysis Techniques
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NSC Recommendations for Conducting a Task Analysis

1. General Observations
 2. Questionnaires and Interviews
 3. Videotaping and Photography
 4. Drawing and Sketching
 5. Measuring the Work Environment
 6. Understanding the Ergonomics of Aging
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Emergency Response Team Personnel

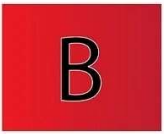
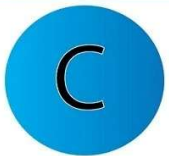
1. Site Main Controller (SMC)
 2. Incident Controller (IC)
 3. Deputy Incident Controller (DIC)
 4. Essential Workmen (EWM)
 5. Other Key Persons
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Requirements of an Emergency Response Plan

An effective ERP should include:

1. Emergency organization/unit.
 2. Scenario vulnerability analysis and consequence assessment.
 3. Emergency Control Centre (ECC).
 4. Sirens, assembly points, evacuation routes, and rescue facilities.
 5. First-aid, medical, and transport facilities.
 6. Emergency declaration and communication system.
 7. Emergency shutdown procedures.
 8. Atmospheric stability and wind velocity information.
 9. Site and area maps.
 10. Roles and responsibilities of emergency personnel.
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MOD 3

		Ordinary Combustibles	Wood, Paper, Cloth, Etc.
		Flammable Liquids	Grease, Oil, Paint, Solvents
		Live Electrical Equipment	Electrical Panel, Motor, Wiring, Etc.
		Combustible Metal	Magnesium, Aluminum, Etc.
		Commercial Cooking Equipment	Cooking Oils, Animal Fats, Vegetable Oils

(i) Effect of Enclosure

1. **Controls Oxygen Supply** – limits oxygen; determines intensity/duration of fire
2. **Traps Combustion Gases** – smoke & hot gases accumulate in upper portion; reduces visibility
3. **Influences Heat Transfer**
 - a) **Convection** – hot gases rise and spread heat
 - b) **Conduction** – heat travels through materials (e.g., metal structures)
 - c) **Radiation** – heat radiated via electromagnetic waves to nearby objects
4. **Types of Fire Based on Enclosure Conditions**
 - **Fuel-Controlled Fire** – sufficient oxygen available; burns until fuel exhausted
 - **Ventilation-Controlled Fire** – insufficient oxygen; growth slows due to lack of ventilation

(ii) Early Detection of Fire

1. **Thermal Expansion Detector** – heat-sensitive fusible link melts → completes circuit → alarm
 2. **Photoelectric Fire Sensor** – smoke interrupts photoelectric beam → relay closes → alarm
 3. **Radiation / Ionization Sensor** – radioactive substance ionizes air; smoke changes ionization → alarm
 4. **UV or Infrared Detector** – detects UV/IR radiation from flames; rapid changes trigger alarm
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Q3. Methods of Extinguishing Fire

1. **Cooling** – removes heat; water absorbs heat, converts to steam, drops temperature below ignition point → Class A fires
 2. **Smothering** – removes oxygen; foam/CO₂/sand blanket cuts off oxygen supply → flammable liquid/gas fires
 3. **Starvation (Removal of Fuel)** – removes fuel source; close valves, firebreaks, remove combustibles → gas/industrial fires
 4. **Chain Breaking (Chemical Inhibition)** – interrupts chemical chain reaction; free radicals neutralised → DCP/Halon extinguishers
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Early Fire Detection Devices

1. Thermal Expansion Detector – detects: Heat
 2. Photoelectric Fire Sensor – detects: Smoke
 3. Radiation / Ionization Sensor – detects: Smoke particles
 4. UV / Infrared Detector – detects: Flame radiation
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Types of fire Extinguisher

1. Water Type Fire Extinguisher
 2. Foam Type Fire Extinguisher
 3. Dry Chemical Powder (DCP) Fire Extinguisher
 4. Carbon Dioxide (CO₂) Fire Extinguisher
 5. Halon Type Fire Extinguisher
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Fire Prevention Activities

1. **Isolation of Fire Triangle Elements** – prevent contact between fuel, oxygen, and heat
2. **Proper Storage of Flammable Liquids** – flame-resistant buildings; tanks away from work areas; underground storage preferred
3. **Use of Less Flammable Materials** – replace highly flammable materials with safer alternatives
4. **Use of Catalysts and Fire Inhibitors** – create endothermic reactions that absorb heat
5. **Elimination of Ignition Sources** – no smoking near combustibles; no open flames near fuels; non-sparking tools
6. **Immediate Cleaning of Flammable Spills** – dispose of contaminated materials safely
7. **Good Housekeeping** – remove unnecessary combustibles; no accumulation of waste paper/cloth/chemicals
8. **Safe Electrical Practices** – cords along walls; prevent damage to wiring

9. **De-energize Equipment Before Maintenance** – isolate power before maintenance work
 10. **Routine Inspection of Fire Extinguishers** – regularly inspect and test
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Causes of Fire at Homes

1. Electrical Faults – damaged wiring, short circuits, overloaded sockets, faulty appliances
 2. Cooking Accidents – overheated oil, unattended stoves, LPG leakage, flammable spills
 3. Smoking Materials – carelessly discarded cigarettes, lit matches/lighters near combustibles
 4. Open Flames – candles, oil lamps, incense sticks, firecrackers
 5. Storage of Flammable Materials – petrol, kerosene, paints, solvents, LPG cylinders
 6. Children Playing with Fire – matches, lighters, firecrackers
 7. Poor Housekeeping – accumulation of paper, cloth, wood, combustible waste
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Fire Preventive Measures for Homes

1. Proper Storage of Flammable Materials – away from heat/ignition sources
 2. Eliminate Ignition Sources – no smoking near combustibles; use non-sparking tools
 3. Maintain Electrical Safety – avoid overloading; replace damaged wires; periodic inspection
 4. Safe Cooking Practices – never leave cooking unattended; check LPG connections
 5. Clean Up Flammable Spills Immediately – dispose cleaning materials safely
 6. Good Housekeeping – dispose waste paper, cloth, packaging materials
 7. Keep Electrical Cords Properly Arranged – along walls; no exposed/damaged cables
 8. Switch Off Power Before Maintenance – de-energize equipment before repair
 9. Install Fire Detection Devices – smoke detectors, heat detectors, photoelectric sensors
 10. Keep Fire Extinguishers Ready – suitable type; inspect and test periodically
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Dos

1. Raise the alarm immediately – activate nearest fire alarm; inform fire control room
2. Remain calm – think clearly; follow emergency procedures
3. Evacuate through staircases – use designated emergency exits
4. Follow emergency instructions – obey fire wardens, security, firefighters
5. Stay low if smoke is present – crawl near floor where air is cleaner
6. Close doors behind you – slows spread of fire and smoke
7. Assist children, elderly, and disabled persons
8. Use fire extinguishers for small fires – only if trained and fire is controllable
9. Check doors before opening – if hot, do not open; use alternate route
10. Assemble at designated safe area – wait for further instructions

Don'ts

1. Do NOT use elevators/lifts – power failure risk; smoke enters lift shafts
2. Do NOT panic – causes stampedes and injuries
3. Do NOT re-enter the building – not until authorities declare it safe
4. Do NOT open hot doors – exposes to flames and intense heat
5. Do NOT hide – never hide in rooms, washrooms, or cupboards
6. Do NOT break windows unnecessarily – increases O₂ supply, intensifying fire
7. Do NOT run – causes falls and crowd congestion

8. Do NOT use water on electrical fires – risk of electrocution
 9. Do NOT block escape routes – keep corridors, staircases, exits clear
 10. Do NOT ignore fire alarms – treat every alarm as a real emergency
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Precautions During a Fire Emergency

1. Raise the alarm immediately – activate alarm; inform fire department
 2. Remain calm – avoid panic; act responsibly
 3. Evacuate immediately – leave through nearest emergency exit
 4. Use staircases only – do NOT use lifts
 5. Stay low in smoke – crawl/bend low; cleaner air near floor
 6. Close doors and windows – prevent spread of fire and smoke
 7. Do not collect personal belongings – leave immediately
 8. Help vulnerable persons – children, elderly, differently-abled
 9. Use fire extinguishers if trained – only for small, controllable fires
 10. Assemble at safe location – designated assembly point
 11. Avoid electrical hazards – no water on electrical fires; isolate power if safe
 12. Follow emergency team instructions – cooperate with firefighters
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Fire Exit Plan for DSATM / Educational Institution

1. Step 1 – Detection & Alarm: Smoke/heat detectors and alarms in classrooms, labs, library, corridors; audible throughout campus
 2. Step 2 – Emergency Notification: Security informs fire dept., administration, emergency response team
 3. Step 3 – Evacuation: Ground floor → nearest emergency doors; Upper floors → emergency staircases + illuminated exit signs; NO lifts
 4. Step 4 – Assembly Point: Designated open area (main playground / open parking / sports ground)
 5. Step 5 – Roll Call & Accountability: Faculty takes attendance; report missing persons to rescue personnel
 6. Step 6 – Firefighting Measures: Extinguishers in classrooms, labs, library, computer centres, electrical rooms; fire hydrants on campus
 7. Step 7 – Emergency Response Team: Fire wardens + security + faculty coordinators + first-aid; guide evacuation, assist injured, coordinate with firefighters
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Importance of Fire Triangle

1. Helps Understand Fire Formation – fire needs all three elements simultaneously
 2. Basis of Fire Prevention – controlling any one element prevents fire
 3. Basis of Fire Extinguishment
 - d) Cooling → removes Heat (Water extinguisher)
 - e) Smothering → removes Oxygen (Foam/CO₂)
 - f) Starvation → removes Fuel (close valves, remove combustibles)
 4. Helps Select Proper Firefighting Methods – different agents remove different elements
 5. Improves Fire Safety Awareness – educates workers on hazards and prevention techniques
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Management Measures(Electrical Hazzard)

1. Overload Protection
 - g) Fuse – melts under excessive current; disconnects circuit to prevent overheating
 - h) Circuit Breaker – trips automatically when current exceeds set value; de-energises circuit
 2. Grounding (Earthing) – low-resistance path for fault current to flow safely to earth; prevents shock and fire
 3. Ground Fault Circuit Interrupter (GFCI) – detects leakage current; auto-disconnects power; provides better protection than earthing alone
 4. Regular Inspection & Maintenance – inspect wiring; replace damaged cables/plugs; maintain insulation
 5. Safe Work Practices – switch off before maintenance; use insulated tools; avoid wet hands; lockout/tagout procedures
 6. Proper Installation – standard components; correct wiring; avoid overloading sockets
 7. Training and Awareness – train workers on hazards; display safety instructions
 8. Use of PPE – insulated gloves, safety shoes, face shields, protective clothing
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Q13. Importance of Electrical Safety

1. Prevents Electric Shock – protects from accidental contact with live parts
 2. Prevents Electrical Fires – controls faulty wiring, overloads, and short circuits
 3. Protects Human Life – prevents muscle contractions, respiratory failure, heart failure, fatalities
 4. Prevents Electrical Burns – deep, painful, slow-healing burns prevented by safe handling
 5. Protects Equipment and Property – fuses/circuit breakers prevent damage to motors, computers, appliances
 6. Ensures Continuous Operation – safe systems reduce unexpected breakdowns; improves productivity
 7. Reduces Financial Losses – avoids repair, replacement, medical, and property costs
 8. Improves Workplace Safety – safer environment; increases employee confidence; reduces accident rates
 9. Compliance with Safety Regulations – meets legal requirements; reduces penalties and legal liabilities
 10. Protects Against Ground Faults and Overloads – through Fuses, Circuit Breakers, and GFCI
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Q14. Technical Requirements of Product Safety Programme

1. Design Review

- i) Stages: Conceptual → Intermediate → Final Pre-production
- j) Design review committee: production, QC, consumer service, management
- k) Preliminary Hazard Analysis (PHA) – identifies hazards before production
- l) Conceptual stage: function, performance, cost, life expectancy, safety, user instructions
- m) Intermediate stage: layout, electrical schematics, specifications, tooling
- n) Final stage: detailed drawings, material specs, manufacturing processes, tolerances

2. Documentation and Change Control

- a) Maintain complete technical records of product development
- b) Documents serve as evidence during litigation
- c) Any change in design/production/distribution must be updated in all documents
- d) Drawings, tests, inspections, quality records must always be current

3. Purchase Product Control

- a) Purchase documents clearly specify design, material, and safety requirements
- b) Select proven and reliable vendors
- c) Verify that supplied materials conform to specifications
- d) Take corrective actions when required; suppliers must report component hazards

4. Manufacturing and Quality Control

- a) Machinery and procedures must comply with safety regulations
- b) Workers trained and certified for critical operations (welding, brazing)
- c) Quality control: inspection, testing, random sampling, statistical analysis

5. Packaging and Marketing

- a) Packaging protects products during storage and transportation
 - b) Must include: assembly instructions, operating instructions, maintenance guidelines, safety warnings
 - c) Warnings required for hazards that cannot be completely eliminated
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Effects of Electricity on Human Body

- 1. Slight Sensation / Tingling – mild discomfort; small currents; usually not serious
 - 2. Painful Muscle Contractions – involuntary contraction; person unable to release conductor
 - 3. Electrical Burns – deep, painful, slow to heal; damages skin, muscles, nerves, internal tissues
 - 4. Respiratory Failure – current affects respiratory muscles; breathing stops
 - 5. Heart Disturbance – irregular heartbeat / cardiac arrest
 - 6. Death – from respiratory failure, heart failure, or fatal burns
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Main Electrical Hazards

1. Electric Shock – contact with live conductor; injury, burns, or death
 2. Electrical Burns – direct contact or arc; generally severe and deep
 3. Fire Hazards – faulty equipment, damaged wiring, short circuits, overloaded sockets
 4. Short Circuit – unintended low-resistance path; excessive heat and sparks
 5. Overloading – excess current causes overheating; damages equipment and starts fires
 6. Arc Flash and Sparks – intense heat; ignites nearby combustible materials
 7. Faulty Grounding – improper earthing exposes users to dangerous voltages; shock risk
 8. Equipment Failure – damaged insulation or defective equipment creates hazardous conditions
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Q16. Safety Measures for Electric Work

1. Switch Off Power Before Working – de-energize equipment; disconnect power completely
 2. Use Proper Grounding (Earthing) – safe path for fault current; prevents electric shock
 3. Install Overload Protection Devices – fuses and circuit breakers disconnect on excess current
 4. Use Ground Fault Circuit Interrupters (GFCI) – detects leakage current; auto-disconnects power
 5. Use Insulated Tools and Equipment – reduces accidental contact with live parts
 6. Wear Personal Protective Equipment (PPE) – insulated gloves, safety shoes, protective clothing, face shields
 7. Inspect Electrical Equipment Regularly – check cables, plugs, switches, insulation; replace defectives
 8. Avoid Overloading Circuits – do not connect too many appliances to one socket
 9. Keep Work Area Dry – never handle equipment with wet hands; avoid damp conditions
 10. Provide Training and Awareness – train workers; first-aid and emergency response procedures
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Q17. Product Safety and Product Safety Programme

Product Safety

- A quality product meets or exceeds customer requirements without causing injury, illness, death, or property damage
- Governed by Product Liability Law and Consumer Product Safety Act (CPSA), 1972
- Seller of defective/unreasonably dangerous product is liable for injuries caused

Objectives of CPSA, 1972

1. Protect the public from unreasonable risks of injury from consumer products
2. Assist consumers in evaluating comparative safety of products
3. Promote research into causes and prevention of product-related deaths, illnesses, and injuries

Product Safety Programme – 5 Technical Requirements

Design Review – Conceptual → Intermediate → Final stages

PHA conducted; review committee evaluates safety

1. Documentation & Change Control – maintain records; update all changes; current drawings/tests
 2. Purchase Product Control – clear purchase specs; verified vendors; corrective actions; suppliers report hazards
 3. Manufacturing & Quality Control – certified workers; inspection, testing, random sampling, statistical analysis
 4. Packaging & Marketing – assembly/operating/maintenance instructions + safety warnings on packaging
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